

- 1 (a) Define gravitational potential at a point.

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..... [2]

- (b) Mars is a planet that may be considered to be an isolated uniform sphere of radius 3.4×10^6 m.

A satellite of mass 122 kg is in orbit around Mars at a constant height of 1.7×10^6 m above the surface of the planet.

The height of the orbit is increased to 6.8×10^6 m above the surface. This increases the gravitational potential energy of the satellite by 5.1×10^8 J.

- (i) Show that the mass of Mars is 6.4×10^{23} kg.

[3]

- (ii) Calculate the gravitational potential ϕ at the surface of Mars. Give a unit with your answer.

$\phi =$ unit [2]

